

PROPHET TIME SERIES FORECASTING ASSIGNMENT

E-Commerce Sales Forecasting

****Duration:**** 2-3 hours

ASSIGNMENT OVERVIEW

You are a data analyst for an e-commerce company. Your manager needs to forecast sales for Q1 2024 (90 days) to plan inventory and staffing.

Your task:

1. Explore the sales data
2. Build a Prophet forecasting model
3. Forecast Q1 2024 sales
4. Answer 4 business questions

DATASET

****File:**** `sales\_data.csv`

Columns:

- `date` - Date of observation
- `sales` - Daily sales revenue in USD
- `ads\_spend` - Daily advertising budget
- `discount` - Daily discount percentage
- `day\_of\_week` - 0-6 (Monday=0, Sunday=6)
- `month` - Month number (1-12)
- `day` - Day of month (1-31)

****Important:**** You must rename the columns:

- Rename `date` to `ds`
- Rename `sales` to `y`

Prophet requires the date column to be named `ds` and the target column to be named `y`.

****Dataset:**** 730 days (2 years) from 2022-01-01 to 2023-12-31

PART 1: DATA EXPLORATION

Task 1.1: Load and Summarize Data

Answer these questions:

1. How many days of data do we have?
2. What is the date range?
3. What is the average daily sales?
4. What is the highest and lowest daily sales?

Task 1.2: Create Visualizations

Create 2 plots:

1. ****Time series plot**** - Show sales over all 730 days
 - Title: "Daily Sales Over Time"
2. ****Day of week comparison**** - Show average sales for each day (Mon-Sun)
 - Title: "Average Sales by Day of Week"

Analyze:

- What pattern do you see in the data?
- Which day of the week has the highest sales?

PART 2: BUILD PROPHET MODEL

Task 2.1: Prepare Data

1. Load the CSV file
2. Rename columns: `date` → `ds`, `sales` → `y`
3. Ensure `ds` is datetime format
4. Split data into training and testing:
 - ****Training:**** Jan 2022 - Oct 2023 (80%)
 - ****Testing:**** Nov-Dec 2023 (20%)

Task 2.2: Build Prophet Model

Build a basic Prophet model with:

- Yearly seasonality enabled
- Weekly seasonality enabled
- Daily seasonality disabled

Do NOT add external regressors (keep it simple).

Task 2.3: Test the Model

1. Make predictions on the test set (Nov-Dec 2023)
2. Calculate MAPE (Mean Absolute Percentage Error)

3. Report the MAPE score

****Interpretation:**** MAPE < 10% is good accuracy.

PART 3: FORECAST Q1 2024

Task 3.1: Generate 90-Day Forecast

Create a forecast for Q1 2024 (January 1 - March 31, 2024).

Report:

- Total expected sales for Q1
- Average daily sales
- Peak day (highest forecast)
- Low day (lowest forecast)

Task 3.2: Save Forecast

Save your 90-day forecast to a CSV file with columns:

- Date
- Forecasted Sales

PART 4: ANSWER BUSINESS QUESTIONS

Question 1: Inventory Planning

****Question:**** "We need to order inventory for Q1. Based on your forecast, how much total sales should we expect? If each product has an average value of \$20, how many units might we sell?"

Answer:

- Total Q1 sales forecast: \$ _____
- Units to sell (at \$20 each): _____ units
- Recommended inventory level: _____ units

Question 2: Which Days are Busiest?

****Question:**** "Which days of the week are busiest in our forecast? How much higher are peak days vs. slow days?"

Answer:

- Busiest days: _____ (day names)
- Slowest days: _____ (day names)
- Peak vs slow ratio: _____ times higher

Question 3: Marketing Analysis

Question: "Looking at the historical data, does higher advertising spend correlate with higher sales? Should we increase our ad budget?"

Answer:

- Correlation between ads and sales: _____ (range: -1 to +1)
- What does this correlation tell you?
- Recommendation: Keep same / Increase / Decrease (choose one)
- Reason: _____

Question 4: Executive Summary

Question: "Write a brief 3-4 sentence summary of your findings for the manager."

Answer:

SUBMISSION CHECKLIST

Submit these files:

1. `forecast\_analysis.ipynb` - Your Jupyter notebook with all code and analysis
2. `sales\_exploration.png` - Time series plot
3. `day\_of\_week.png` - Day of week bar chart
4. `q1\_2024\_forecast.csv` - 90-day forecast data (date + forecasted sales)
5. `ANSWERS.txt` - Your answers to the 4 business questions